ALEJANDRO GARCIA SORIA

agarciaso02@gmail.com ⋄ +34 692 583 861
agarciasoria.github.io

PROFILE

Theoretical physicist with dual undergraduate degrees in Physics and Mathematics and an M.Sc. in Theoretical Physics from the Instituto de Física Teórica (IFT, CSIC–UAM). My research background revolves around AdS/CFT correspondence, higher-derivative gravity, and strongly coupled quantum field theory. I am seeking PhD training in gravity, quantum field theory, and the AdS/CFT correspondence.

RESEARCH INTERESTS

- AdS/CFT correspondence and gauge/gravity duality
- Gravitational effective field theories and higher–derivative gravity
- Strongly coupled quantum field theory and holographic transport
- Emergence of low–energy physics from effective theories

EDUCATION

Autonomous University of Madrid (UAM) 2024 – 2025
M.Sc. in Theoretical Physics (IFT–CSIC)
Specialisation in particle physics and cosmology.
Master's thesis in holography, gravity, and relativistic hydrodynamics.

University of Oviedo 2018 – 2023
B.Sc. in Physics
Bachelor's theses in General Relavitity: Black Holes and Wormholes.

University of Oviedo 2018 – 2023
B.Sc. in Mathematics
Bachelor's theses in Dynamical Systems: Slow passage through a Hopf Bifurcation

RESEARCH EXPERIENCE

Instituto de Física Teórica (IFT–CSIC) 2024 – 2025
Master's Thesis: Higher–Derivative Gravity and Holographic QCD *Madrid, Spain*

· Investigated strongly coupled quantum field theories using gravitational duals in asymptotically AdS spacetimes.
· Derived a general analytic expression for the shear viscosity ratio η/s in five–dimensional dilatonic quasi–topological gravity, extending existing results beyond perturbative curvature expansions.
· Solved coupled black hole background equations numerically to extract thermodynamic and transport properties of the dual field theory.
· Analyzed the impact of higher–curvature and scalar couplings on conformal symmetry breaking and non–universal transport behavior.
· Gained experience with effective gravitational actions, holographic renormalization, and bulk–boundary dynamics.

JAE Intro – CSIC Oct 2024 – Feb 2025
Research Intern in Theoretical Physics *Madrid, Spain*

· Competitive research fellowship funded by the Spanish National Research Council (CSIC).
· Conducted research on gravity duals of strongly coupled quantum systems within the framework of holography.

· Applied symbolic and numerical methods to analyze coupled bulk field equations in higher–derivative gravity models.

ACADEMIC PROJECTS

Bachelor's Thesis – Physics 2023
General Relativity and exotic solutions *University of Oviedo*

· Analytical study of exotic exact solutions of Einstein's equations such as Black Holes and Wormholes and their geometric and physical properties.

Bachelor's Thesis – Mathematics 2023
Dynamical Systems and Bifurcation Theory *University of Oviedo*

· Rigorous analysis of nonlinear dynamical systems, stability properties, and Hopf bifurcations.

AWARDS & FUNDING

- JAE Intro Research Fellowship (CSIC, 2024)
- Academic Excellence Scholarship for Master's Studies (UAM, 2024)
- Silver Medal – Regional Physics Olympiad (La Rioja, 2018)

LANGUAGES

Spanish (Native)
English (C2 – Cambridge Certified)
French (Basic)

PERSONAL REFERENCES

- Professor Juan F. Pedraza, Institute for Theoretical Physics (UAM-CSIC), email: j.pedraza@csic.es
- Professor Hyun-Sik Jeong, Asia Pacific Center for Theoretical Physics , email: hyunsik@postech.ac.kr